

Education

Bachelor of Software Engineering (B. Seng) | University of Victoria

Sept 2024 – Dec 2028

Relevant courses – Data Structures & Algorithms, Software Development Methods, Object Oriented Programming, Fundamentals of Programming (C, Java & Python). **GPA: 4.0 / 4.0**

Technical Skills

Programming Languages: Python, C, Java, JavaScript, TypeScript, bash, SQL (MySQL), MATLAB, Thonny, R

Web development: HTML, CSS, Embedded JavaScript (EJS), RESTful APIs.

Frameworks & Libraries: Flask, FastAPI, Node.JS, React.js, Express.js

Databases & Data Tools: MySQL, PostgreSQL, SQLite, Microsoft Excel, Power BI, Microsoft Automate

Language Engineering & Compilers: Lark (LALR), grammar design (EBNF), AST design, visitor / Interpreter pattern

Tools & Platforms: Git/GitHub, Docker, Playwright, VS Code, Jupyter Notebook, Markdown

Testing & QA: Unit Testing, Pytest, Debugging, parse-tree inspection, Data Validation, Fault-Tolerant Design

Embedded Systems & IoT: Raspberry Pi Pico, MicroPython, GPIO, Real-Time Data Acquisition.

Other: Agile Methodologies, Technical Documentation, Data Visualization

Technical Projects

Research Assistant | Solids Lab, UVIC

Sep 2025 - Present

- Practicing **ethical attacks** in lab by using **Kali, Nmap, Burp** on vulnerable apps so that I learn, exploitation basics.
- Building detections by streaming **Linux / web logs** into Elastic and writing simple **queries** so that brute force & scans are flagged.
- Learning **OWASP Top-10** by reproducing **XSS / SQLi** and documenting fixes so that I can explain root cause & remediation.
- Hardening **Ubuntu VMs** by tightening **SSH**, least-privilege, and firewall rules so that common paths are blocked before tests.

Backend Developer | University of Victoria Satellite Design Club, UVIC ([Github](#))

Jan 2025 - Present

- Developed a **Python-based scalable backend** using traceable, **fault-tolerant architecture** to **automate** handling of binary image / telemetry data from satellite systems, reducing data loss by **40%**.
- Automated **data reassembly** and **segmentation** via **file I/O** and control structures to ensure consistent **identifier tagging** across sessions.
- Developed a **pytest test suite** with edge-case coverage to increase backend stability, cutting regressions by **50%**.
- Documented test protocols and quality checks using clear **technical writing** to streamline cross-team **maintenance**.
- Collaborated with teams via **GitHub** and **agile practices** to resolve conflicts and accelerate feature deployment.
- Identified and **debugged edge cases** in live **data capture** to ensure **99% uptime** during critical missions.

DevSwipe – GitHub Discovery Platform | Personal Project ([Github](#))

Sep 2025 - Present

- Building a Tinder-style **web app** with **Flask, SQLAlchemy**, and **GitHub API** to simplify repository discovery.
- Implementing authentication via **Werkzeug hashing** and sessions to maintain **100% data integrity** and **security**.
- Designing an **SQLite schema** with **1-to-many relationships** to enable personalized **user tracking** and swipes.
- Enhancing engagement with **swipe mechanics** and filters to boost average session time by **30%**.

Full Stack Developer-Smart Fall Prevention System | University of Victoria (Design Lab)

Jan 2025 – Apr 2025

- Developed a **Flask-based full-stack** system with **Raspberry Pi Pico** and **MicroPython** to detect falls in real time, reducing alert delays by **40%**.
- Integrated live sensor data via **GPIO** and **ADC modules** to trigger instant emergency notifications, improving response time by **50%**.
- Designed a dynamic dashboard using **JavaScript** to visualize motion trends, enhancing caregiver monitoring efficiency.

Hackathons

University of Victoria Engineering Competition - Programming (UVEC) ([Github](#))

Oct 2025

- Used **Python + LALR grammar** to design a **typed mini-language** with expressions, conditionals, and loops and delivered a working student-friendly language demo in **6 hours**.
- Used a **scoped runtime environment** to manage variable bindings and short-circuit logic and delivered reliable execution with clear beginner-friendly errors.
- Used a **visitor-based evaluator** to implement correct while / for control flow and delivered stable behavior under live demo conditions.
- Used a rapid **debug** workflow (**parse-tree inspection** and sanity tests) and delivered fast iteration with consistent, reproducible results.
- Used structured **team planning** and responsible AI scaffolding to split **grammar / evaluator / CLI** work and delivered an **on-time MVP** and **confident presentation**.

Professional Experience

Self-Checkout Assistant / Cashier | Walmart, Victoria, BC.

Sept 2025 - Present

- **Support customers** at self-checkout stations by assisting with transactions, **troubleshooting issues**, and ensuring a smooth shopping experience.
- **Monitor registers** and verify purchases to maintain compliance with store policies and regulations.
- **Maintain efficiency** by quickly resolving machine errors and escalating security concerns to supervisors.

Voting Information Officer | UVSS, University of Victoria, Canada.

March 2025

- Maintained **100%** accurate voter records for **5,000+ students** using **Excel** and UVSS policies.
- Resolved eligibility issues and supported **smooth participation** during peak election periods.
- Collaborated with staff under tight deadlines to ensure **transparency** and **integrity**.

Certifications

- **Introduction to Cybersecurity** — Cisco Networking Academy, (Planned Sept 2025)
- **CompTIA Cybersecurity Fundamentals** — CompTIA, (Planned Sept 2025)
- **Prompt Engineering for AI** — Udemy, Ongoing
- **Python for Everybody** — University of Michigan (Coursera), May 2024
- **CS50: Introduction to Computer Science** — Harvard, June 2023

Scholarships & Awards

- **Certificate of Achievement** – UVic Math & Stats Dept. (2025)
- **Entrance Scholarship for Academic Excellence** – UVic (2024)

Extracurricular Activities

- Volunteer Educator with **PRAYAS NGO** since 2019, **teaching** underprivileged children and reinforcing the importance of **accessibility in learning**.
- Write and perform original poetry, building **creative expression**, **public speaking**, and **emotional intelligence**.
- Play drums and guitar as a self-taught musician, developing **discipline**, **coordination**, and creative **collaboration**.
- Edit personal videos and event montages, showcasing storytelling, **attention to detail**, and digital content skills.
- Compete in cricket, squash, and swimming, strengthening **teamwork**, **focus**, and **resilience** through sports.
- Learning **French** to enhance **bilingual communication** skills and align with Canada's official language **culture**.